

An Open, Zero-Rent Protocol for Distributed Edge Compute and Authorized Bandwidth Relaying

Abstract

Centralized machine learning (ML) infrastructure and commercial data-relaying networks face high capital expenditure, middleman rent-seeking, and rigid deployment topologies. This paper proposes an open, stateless, zero-rent protocol designed to orchestrate volunteer edge compute and authorized network relaying over heterogeneous legacy hardware. By eliminating native protocol tokens, middleman fees, and centralized control layers, the system enables direct peer-to-peer (P2P) coordination. We outline a modular architecture featuring localized, pre-loaded ML inference, probabilistic result verification, authorization-first network routing, and a concrete threat model for host safety and protocol integrity.

1. Introduction and Economic Architecture

1.1 Capital Expenditure vs. Operational Expenditure

While modern data center accelerators offer unmatched compute density per watt, acquiring new hardware requires substantial capital expenditure (CapEx). Conversely, legacy and idle consumer hardware (e.g., decommissioned laptops, single-board computers) carries zero marginal CapEx, operating purely on local operational expenditure (OpEx / electricity).

This protocol provides an open orchestration layer to aggregate idle, heterogeneous hardware into a functional edge compute network. Operators ("scavengers") monetize underutilized hardware and household bandwidth up to the point where local electricity costs match market compute yields.

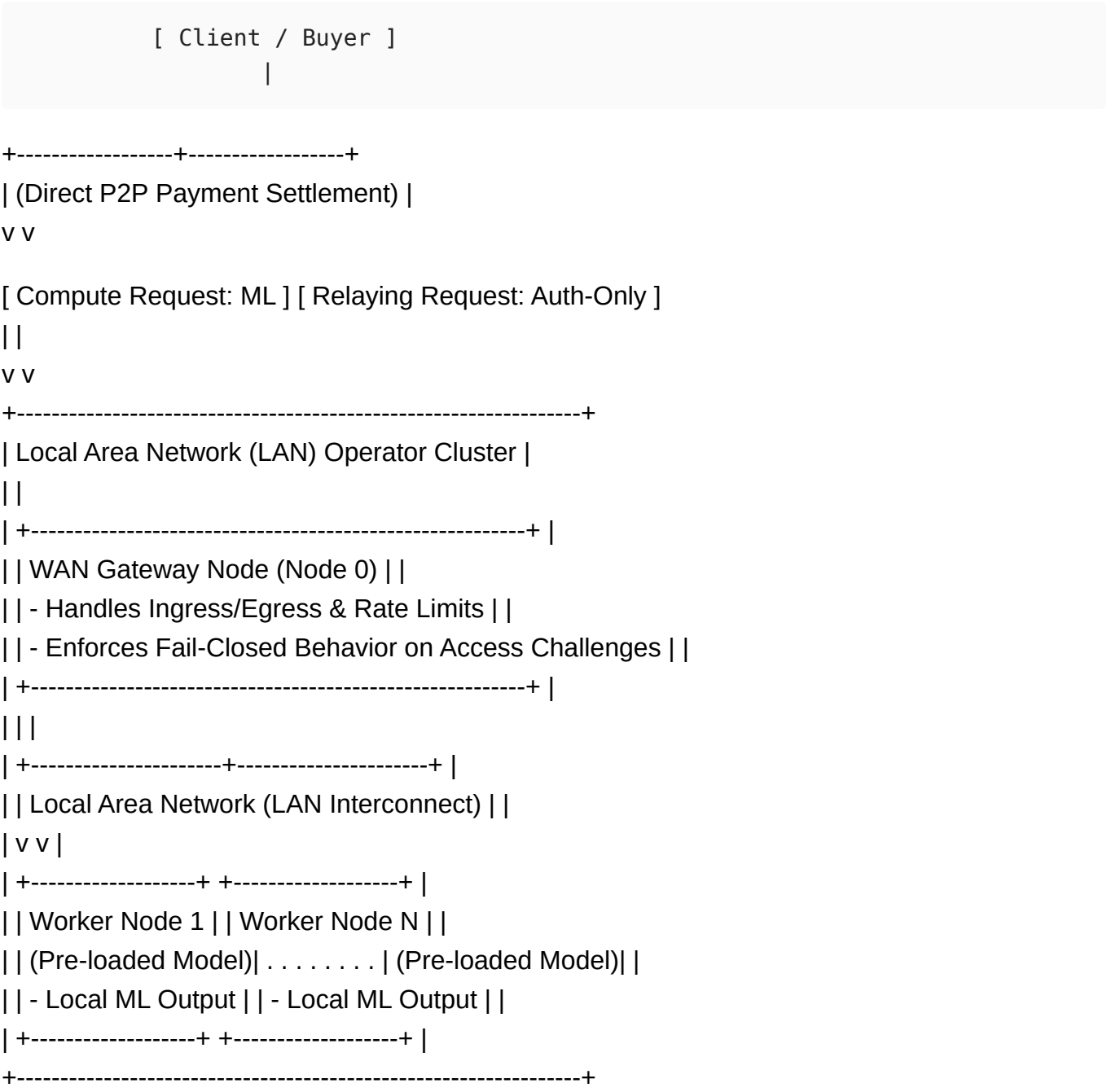
1.2 Zero-Rent Protocol Mechanics

To minimize operational friction and avoid speculative financial overhead:

- **No Native Protocol Token:** Value transfers settle directly between buyers and providers via existing digital payment rails (e.g., Layer-2 stablecoins, state channels, or direct micro-settlement APIs).
 - **Zero Protocol Fee:** 100% of buyer payments route directly to the executing node operator.
 - **Minimal State Overlay:** The network operates purely as an open discovery and verification protocol, maintaining no centralized state or custodial funds.
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2. System Topology and Node Categorization

The network decouples compute tasks from WAN bandwidth capabilities to optimize hardware utilization and respect household network constraints.



2.1 The LAN Node Cluster

Multiple devices operating behind a single public WAN gateway automatically form a local cluster:

1. **Primary Gateway Node:** One node manages public network interactions, enforcing rate limits, local bandwidth caps, and network isolation policies.

2. **Internal Compute Workers:** devices on the local LAN process offloaded compute tasks. Compute nodes do not route raw external proxy traffic, preventing internal network saturation and IP address degradation.
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3. Workload Specification and Output Verification

Large-scale foundation models require high-bandwidth interconnects unsuited for legacy consumer hardware. Therefore, this protocol specifically targets **Localized, Quantized Edge Inference** (e.g., 1B–3B parameter models, text/image classification, feature extraction).

3.1 Pre-Loaded Model Storage

To eliminate the prohibitive overhead of transferring multi-gigabyte model weights over consumer connections, nodes maintain a local library of standard, open-weights models. Tasks specify a deterministic model hash; if a node lacks the required model locally, it rejects the job.

3.2 Probabilistic Verification

Because output hashes only confirm data receipt—not computational correctness—the client software employs a **Probabilistic Sampling Verification** model:

1. **Spot Checks:** The buyer client assigns tasks redundantly to two independent nodes.
 2. **Deterministic Comparison:** For deterministic workloads (e.g., low-temperature LLM generation or feature extraction), outputs and intermediate tensor state hashes must match within an acceptable floating-point tolerance ().
 3. **Reputation Penalties:** Nodes returning inconsistent results are flagged locally by the buyer client, dropping their allocation priority for future high-value tasks.
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4. Authorization-First Network Relaying

To ensure strict legal compliance, operational transparency, and protection for node operators, network routing functionality operates under a strict **Authorization-First Framework**.

4.1 Explicit Destination Permissions

Node operators must explicitly configure allowed routing targets (e.g., specific domain allowlists, internal API endpoints, or partner networks). The protocol strictly prohibits arbitrary, unconstrained open-proxy behavior.

4.2 Fail-Closed Rate Limiting and Challenge Handling

- Rather than attempting to bypass anti-bot measures, evade access controls, or alter traffic signatures:
- **Fail-Closed Principle:** If an outbound request encounters a rate limit (HTTP 429), access challenge, or authorization failure (HTTP 403), the gateway node instantly halts requests to that target destination.
 - **No Evasion Mechanics:** The protocol does not inject artificial human jitter, alter headers to misrepresent user agents, or attempt to mask network origins. Rate limits are treated as explicit instructions to back off or seek formal API access.
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5. Threat Model and Security Framework

Threat Vector	Attack Mechanics	Protocol Mitigation
Malicious Node	Submits dummy ML results to claim payment without compute.	Probabilistic duplicate sampling + client-side reputation tracking.
Malicious Buyer	Attempts to run unsafe arbitrary code or exploit network endpoints.	Containerized sandboxing (WASM/OIC) + strict host permission controls.
Sybil Subnet Claim	Datacenter nodes impersonating residential endpoints.	Hardware attestation + baseline RTT and hop-distance profiling checks.
ISP ToS Violation	High-volume relaying triggering account suspension.	Host-configured bandwidth caps + fail-closed non-residential routing.

5.1 Host Sandboxing and Resource Controls

All workload execution occurs within isolated runtime environments (e.g., WebAssembly micro-runtimes or unprivileged OCI containers). Executing tasks are restricted from accessing local host storage, adjacent LAN devices, or unmapped network interfaces.

5.2 Sybil and Topology Classification

While Autonomous System Number (ASN) lookups distinguish data center subnets from residential ISPs, they do not prove physical location or hardware identity. The protocol combines ASN mapping with Round-Trip Time (RTT) Latency Profiling across multiple peer nodes to establish relative network distance and verify topology claims without requiring centralized identity checks.

6. Economic Realities and Operator Viability

6.1 Cost-Benefit Equilibrium

An operator's net income is defined by:

- **Compute Selection:** Nodes automatically pause task execution if market task yields fall below local electricity costs.
 - **Hardware Lifecycle:** The protocol provides an operational runway for hardware that would otherwise be landfilled, shifting the economic barrier from CapEx hardware acquisition to pure energy efficiency.
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7. Conclusion

This protocol provides a practical, legally defensible framework for distributed edge compute and authorized network relaying. By abandoning rent-seeking crypto-tokens, rejecting anti-detection evasion techniques, and focusing on localized, verifiable ML workloads, the system establishes a clean, open utility layer for volunteer compute infrastructure.